

Performance Testing

Date	27 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API
Maximum Marks	4 Marks

Performance Testing - Benchmarked Sample Calls

The four calls below were captured directly from Postman/Thunder Client against the team's own local MongoDB instance (server run on port 5001 for this session), with response status, time and payload size read from the client for each call.

S.No	Endpoint	Method	Status	Response Time	Response Size
1	/api/auth/register (user "sabari")	POST	201 Created	664 ms	326 Bytes
2	/api/auth/login (user "sabari")	POST	200 OK	115 ms	314 Bytes
3	/api/auth/profile	GET	200 OK	48 ms	232 Bytes
4	/api/ai/answer (question: "what is mongodb")	POST	200 OK	2.56 s	395 Bytes

Observations

- Registration (664 ms) is the slowest MongoDB-only call because it performs a bcrypt hash-and-salt round (`genSalt(10)`) on every request before writing the new user document.
- Login (115 ms) and profile retrieval (48 ms) are both fast, single-document Mongoose lookups with no hashing work on the read path.
- The AI direct-answer call (2.56 s) is over an order of magnitude slower than any database-only call, because it depends on a live round trip to the external Google Gemini API rather than only the Express/MongoDB layer.
- All four calls returned their expected success status (201 for the creating register call, 200 for the remaining three), with no error responses observed during this test session.

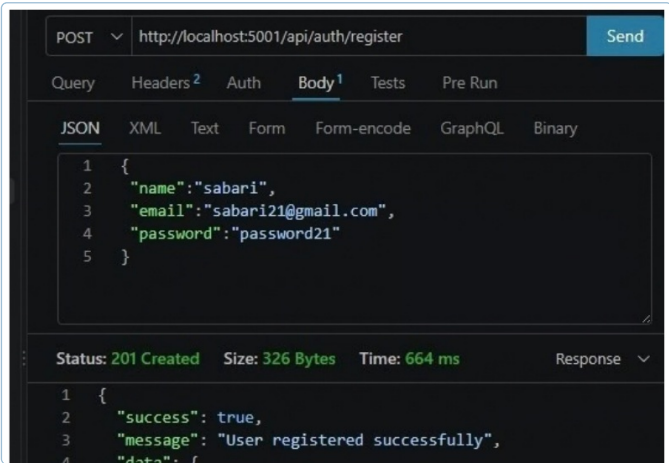
Other Implemented Endpoints (Functionally Verified)

The remaining endpoints below were exercised with real request payloads (see EPIC 5 - Execution & API Testing Evidence in the Project Documentation) and are implemented and routable, but were not captured with an individual response-time screenshot the way the four calls above were.

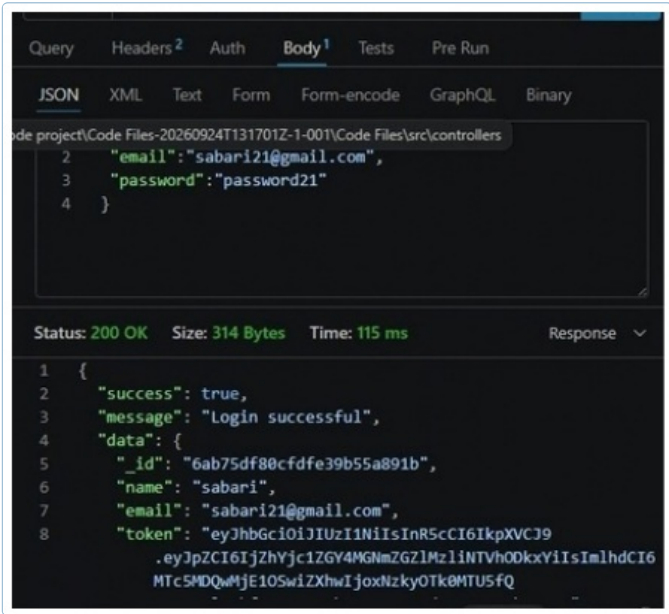
Endpoint	Method	Purpose
/api/faqs	POST	Creates a new FAQ under the authenticated creator
/api/faqs	GET	Lists all FAQ records
/api/faqs/:id	GET / PUT / DELETE	Retrieves, updates or deletes a single FAQ by ID
/api/faqs/search?q=...	GET	Keyword search across FAQ records
/api/ai/generate-faq	POST	Generates a draft FAQ question/answer pair from a topic via Google Gemini

No response time was invented for these five endpoints; a full timing pass across the FAQ CRUD/search routes and the FAQ-generation endpoint is recorded as a next step under Scalability & Future Plan.

Evidence - Sample Test Executions



Register - 201 Created, 326 Bytes, 664 ms



Login - 200 OK, 314 Bytes, 115 ms

